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Editor-in-Chief
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Dear Editor,

We submit for consideration in *Malaria Journal* our manuscript titled “**Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis**” as an original research article.

Manuscript scope and significance

This work addresses a central challenge in antimalarial drug discovery: how to prioritize chemotypes that combine target breadth with mutation resilience while remaining computationally accessible to research groups in endemic regions. We present an integrated workflow that connects African-natural-product-inspired chemical-space expansion to target-anchored docking against four mechanistically distinct *Plasmodium falciparum* proteins (PfDHFR, PfCRT, PfClpP, PfATP4) and per-target resistance-resilience analysis across clinically relevant mutant panels. By keeping these evidence layers separate rather than collapsing them into a single composite score, the workflow generates experimentally testable hypotheses about chemotype utility without conflating computational estimates with biological activity.

The study is particularly relevant to *Malaria Journal*’s scope for several reasons:

1. **Endemic-region research capacity:** The centroid-based library reduction achieves 99.3% computational-cost reduction, making systematic multi-target and mutation-panel screening accessible to groups with limited infrastructure while retaining 69.3% scaffold recovery from African natural-product seeds.
2. **African natural-product chemical space:** The 65 856-molecule hybrid library combines 396 African natural products with 454 synthetic antimalarial compounds, yielding 19 913 computationally prioritized candidates with 92.6% whole-molecule novelty (Tanimoto < 0.4 to seed set).
3. **Mutation-resilience framework:** Per-target resistance-resilience scores (RRS) quantify docking-score retention across PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels, excluding non-binding wild-type baselines to prevent artifactual resilience claims. The framework correctly reports neutral-within-null findings where no mutant signal is detected, strengthening confidence in genuine A* classifications.
4. **Honest-negative results:** The DEKOIS 2.0 near-chance enrichment (PfDHFR AUC 0.45) and the approved-drug retrospective (pyrimethamine and chloroquine RRS indistinguishable from null) constrain interpretation of absolute docking scores while preserving relative within-target rankings—an evidence boundary appropriate for hypothesis-

generation studies.

5. **Experimentally actionable hypotheses:** Five priority candidates (PP-15, PP-05, PP-06, PP-11, PP-13) combine favorable four-target profiles with A* mutation-resilience classification, defining a focused experimental follow-up tier for biochemical assays, mutant-panel validation, and molecular-dynamics refinement.

Contribution to malaria drug discovery

Where recent machine-learning and structure-based approaches have focused on potency prediction from single-target datasets, our workflow prioritizes multi-target engagement and mutation tolerance as distinct, complementary properties. The target panel spans mechanistically unrelated antimalarial vulnerabilities (folate metabolism, chloroquine-resistance transporter, proteostasis, ion homeostasis), testing whether African-natural-product-inspired chemotypes distribute across pathway, transporter, proteostasis, and ion-regulation hypotheses. This breadth aligns with recent experimental polypharmacology screens identifying multi-target engagement as promising for resistance management, while the per-target RRS framework extends conventional resistance-mutation docking by excluding non-binding baselines and preserving target-specific evidence quality.

Limitations and scope

We emphasize that this is a hypothesis-generating study. Docking scores are empirical scoring-function outputs calibrated neither across heterogeneous binding sites nor against experimental free energies. PfCRT is represented by a corrected proxy cavity, and PfATP4 by mechanism-motivated regions without co-crystallized inhibitors. No IC₅₀/EC₅₀ measurements were performed. The resulting candidate profiles propose which chemotypes, targets, and mutant panels should be tested next; they do not claim biological potency, target engagement, or resistance circumvention. This scope is appropriate for computational antimalarial discovery and aligns with *Malaria Journal*'s publication of method-development and hypothesis-generation studies that advance accessible, African-centered research capacity.

Data availability and reproducibility

All data, code, machine-readable tables, raw docking outputs, and provenance records are archived on Zenodo under an MIT license at <https://doi.org/10.5281/zenodo.22696778>. The deposit includes file-level checksum manifests, enabling full computational reproduction.

Article Processing Charge (APC) waiver request

The corresponding author is based in Cameroon, a country classified under BioMed Central's APC waiver policy. We respectfully request a full APC waiver for this submission. All computational work was conducted using institutional resources; no specific external funding was available for this study. An APC waiver would enable dissemination of this work through *Malaria Journal*'s open-access platform, maximizing accessibility to researchers in endemic regions who may benefit from the centroid-based library reduction and per-target mutation-resilience framework.

Author contributions and competing interests

M.V.S.T. and S.G.N.E. designed the integrated study. M.V.S.T. and J.-P.T.N. developed the chemical-space and docking analyses. P.S. and W.F.M. contributed to target biology and resistance interpretation. All authors reviewed and approved the manuscript. We declare no competing financial interests.

Why Malaria Journal?

We chose *Malaria Journal* for its unambiguous scope match (antimalarial discovery, resistance, African research capacity), fast decision timeline (median 9 days to first decision), strong impact factor (3.7), and commitment to supporting endemic-region research through APC waivers. After two desk rejections from journals requiring methodological novelty as a gate, *Malaria Journal*'s focus on biological and public-health significance provides the appropriate venue for this hypothesis-generation study.

We believe this work will interest *Malaria Journal*'s readership—particularly researchers in endemic regions seeking accessible computational workflows, groups exploring African-natural-product chemical space, and experimentalists prioritizing multi-target, mutation-resilient candidates for follow-up assays. We look forward to your editorial decision and, if appropriate, constructive peer review.

Sincerely,

Myke Vital Sao Temgoua
On behalf of all authors